

ELEVEN

"I TAKE YOUR POINT"

Entering Class Discussions



HAVE YOU EVER been in a class discussion that feels less like a genuine meeting of the minds than like a series of discrete, disconnected monologues? You make a comment, say, that seems provocative to you, but the classmate who speaks after you makes no reference to what you said, instead going off in an entirely different direction. Then, the classmate who speaks next makes no reference either to you or to any one else, making it seem as if everyone in the conversation is more interested in their own ideas than in actually conversing with anyone else.

We like to think that the principles this book advances can help improve class discussions, which increasingly include various forms of online communication. Particularly important for class discussion is the point that our own ideas become more cogent and powerful the more responsive we are to others, and the more we frame our claims not in isolation but as responses to what others before us have said. Ultimately, then, a good face-to-face classroom discussion (or online communication) doesn't just happen spontaneously. It requires the same sorts of disciplined moves and practices used in many writing situations, particularly that of identifying to what and to whom you are responding.

FRAME YOUR COMMENTS AS A RESPONSE TO SOMETHING THAT HAS ALREADY BEEN SAID

The single most important thing you need to do when joining a class discussion is to link what you are about to say to something that has already been said.

- ▶ I really liked Aaron's point about the two sides being closer than they seem. I'd add that both seem rather moderate.
- ▶ I take your point, Nadia, that _____. Still . . .
- ▶ Though Sheila and Ryan seem to be at odds about _____, they may actually not be all that far apart.

In framing your comments this way, it is usually best to name both the person and the idea you're responding to. If you name the person alone ("I agree with Aaron because _____"), it may not be clear to listeners what part of what Aaron said you are referring to. Conversely, if you only summarize what Aaron said without naming him, you'll probably leave your classmates wondering whose comments you're referring to.

But won't you sound stilted and deeply redundant in class if you try to restate the point your classmate just made? After all, in the case of the first template above, the entire class will have just heard Aaron's point about the two sides being closer than they seem. Why then would you need to restate it?

We agree that in oral situations, it does often sound artificial to restate what others just said precisely because they just said it. It would be awkward if, on being asked to pass the

salt at lunch, one were to reply: "If I understand you correctly, you have asked me to pass the salt. Yes, I can, and here it is." But in oral discussions about complicated issues that are open to multiple interpretations, we usually do need to resumm-
arize what others have said to make sure that everyone is on the same page. Since Aaron may have made several points when he spoke and may have been followed by other com-
mentators, the class will probably need you to summarize which point of his you are referring to. And even if Aaron made only one point, restating that point is helpful, not only to remind the group what his point was (since some may have missed or forgotten it) but also to make sure that he, you, and others have interpreted his point in the same way.

**TO CHANGE THE SUBJECT,
INDICATE EXPLICITLY THAT YOU ARE DOING SO**

It is fine to try to change the conversation's direction. There's just one catch: you need to make clear to listeners that this is what you are doing. For example:

- ▶ So far we have been talking about the characters in the film. But isn't the real issue here the cinematography?
- ▶ I'd like to change the subject to one that hasn't yet been addressed.

You can try to change the subject without indicating that you are doing so. But you risk that your comment will come across as irrelevant rather than as a thoughtful contribution that moves the conversation forward.

**BE EVEN MORE EXPLICIT
THAN YOU WOULD BE IN WRITING**

Because listeners in an oral discussion can't go back and reread what you just said, they are more easily overloaded than are readers of a print text. For this reason, in a class discussion you will do well to take some extra steps to help listeners follow your train of thought. (1) When you make a comment, limit yourself to one point only though you can elaborate on this point, fleshing it out with examples and evidence. If you feel you must make two points, either unite them under one larger umbrella point, or make one point first and save the other for later. Trying to bundle two or more claims into one comment can result in neither getting the attention it deserves. (2) Use metacommentary to highlight your key point so that listeners can readily grasp it.

- ▶ In other words, what I'm trying to get at here is _____.
- ▶ My point is this: _____.
- ▶ My point, though, is not _____, but _____.
- ▶ This distinction is important because _____.

TWELVE

"WHAT'S MOTIVATING THIS WRITER?"

Reading for the Conversation



“WHAT IS THE AUTHOR’S ARGUMENT? What is he or she trying to say?” For many years, these were the first questions we would ask our classes in a discussion of an assigned reading. The discussion that resulted was often halting, as our students struggled to get a handle on the argument, but eventually, after some awkward silences, the class would come up with something we could all agree was an accurate summary of the author’s main thesis. Even after we’d gotten over that hurdle, however, the discussion would often still seem forced, and would limp along as we all struggled with the question that naturally arose next: Now that we had determined what the author was saying, what did we ourselves have to say?

For a long time we didn’t worry much about these halting discussions, justifying them to ourselves as the predictable result of assigning difficult, challenging readings. Several years ago, however, as we started writing this book and began thinking about writing as the art of entering conversations, we latched onto the idea of leading with some different questions: “What other argument(s) is the writer responding to?” “Is the writer

disagreeing or agreeing with something, and if so what?" "What is motivating the writer's argument?" "Are there other ideas that you have encountered in this class or elsewhere that might be pertinent?" The results were often striking. The discussions that followed tended to be far livelier and to draw in a greater number of students. We were still asking students to look for the main argument, but we were now asking them to see that argument as a response to some other argument that provoked it, gave it a reason for being, and helped all of us see why we should care about it.

What had happened, we realized, was that by changing the opening question, we changed the way our students approached reading, and perhaps the way they thought about academic work in general. Instead of thinking of the argument of a text as an isolated entity, they now thought of that argument as one that responded to and provoked other arguments. Since they were now dealing not with *one* argument but at least *two* (the author's argument and the one[s] he or she was responding to), they now had alternative ways of seeing the topic at hand. This meant that, instead of just trying to understand the view presented by the author, they were more able to question that view intelligently and engage in the type of discussion and debate that is the hallmark of a college education. In our discussions, animated debates often arose between students who found the author's argument convincing and others who were more convinced by the view it was challenging. In the best of these debates, the binary positions would be questioned by other students, who suggested each was too simple, that both might be right or that a third alternative was possible. Still other students might object that the discussion thus far had missed the author's real point and

suggest that we all go back to the text and pay closer attention to what it actually said.

We eventually realized that the move from reading for the author's argument in isolation to reading for how the author's argument is in conversation with the arguments of others helps readers become active, critical readers rather than passive recipients of knowledge. On some level, reading for the conversation is more rigorous and demanding than reading for what one author says. It asks that you determine not only what the author thinks, but how what the author thinks fits with what others think, and ultimately with what you yourself think. Yet on another level, reading this way is a lot simpler and more familiar than reading for the thesis alone, since it returns writing to the familiar, everyday act of communicating with other people about real issues.

DECIPHERING THE CONVERSATION

We suggest, then, that when assigned a reading, you imagine the author not as sitting alone in an empty room hunched over a desk or staring at a screen, but as sitting in a crowded coffee shop talking to others who are making claims that he or she is engaging with. In other words, imagine the author as participating in an ongoing, multisided, conversation in which everyone is trying to persuade others to agree or at least to take his or her position seriously.

The trick in reading for the conversation is to figure out *what views the author is responding to* and *what the author's own argument is*—or, to put it in the terms used in this book, to determine the “they say” and how the author responds to it. One of

the challenges in reading for the "they say" and "I say" can be figuring out which is which, since it may not be obvious when writers are summarizing others and when they are speaking for themselves. Readers need to be alert for any changes in voice that a writer might make, since instead of using explicit road-mapping phrases like "although many believe," authors may simply summarize the view that they want to engage with and indicate only subtly that it is not their own.

Consider again the opening to the selection by David Zinczenko on p. 195.

If ever there were a newspaper headline custom made for Jay Leno's monologue, this was it. Kids taking on McDonald's this week, suing the company for making them fat. Isn't that like middle-aged men suing Porsche for making them get speeding tickets? Whatever happened to personal responsibility?

I tend to sympathize with these portly fast-food patrons, though. Maybe that's because I used to be one of them.

—DAVID ZINCZENKO, "Don't Blame the Eater"

Whenever we teach this passage, some students inevitably assume that Zinczenko must be espousing the view expressed in his first paragraph: that suing McDonald's is ridiculous. When their reading is challenged by their classmates, these students point to the page and reply, "Look. It's right here on the page. This is what Zinczenko wrote. These are his exact words." The assumption these students are making is that if something appears on the page, the author must endorse it. In fact, however, we ventriloquize views that we don't believe in, and may in fact passionately disagree with, all the time. The central clues that Zinczenko disagrees with the view expressed in his opening

See Chapter 6
for more
discussion of
naysayers.

paragraph come in the second paragraph, when he finally offers a first-person declaration and uses a contrastive transition, "though," thereby resolving any questions about where he stands.

WHEN THE "THEY SAY" IS UNSTATED

Another challenge can be identifying the "they say" when it is not explicitly identified. Whereas Zinzchenko offers an up-front summary of the view he is responding to, other writers assume that their readers are so familiar with these views that they need not name or summarize them. In such cases, you the reader have to reconstruct the unstated "they say" that is motivating the text through a process of inference.

See, for instance, if you can reconstruct the position that Tamara Draut is challenging in the opening paragraph of her essay "The Growing College Gap."

"The first in her family to graduate from college." How many times have we heard that phrase, or one like it, used to describe a successful American with a modest background? In today's United States, a four-year degree has become the all-but-official ticket to middle-class security. But if your parents don't have much money or higher education in their own right, the road to college—and beyond—looks increasingly treacherous. Despite a sharp increase in the proportion of high school graduates going on to some form of postsecondary education, socio-economic status continues to exert a powerful influence on college admission and completion; in fact, gaps in enrollment by class and race, after declining in the 1960s and 1970s, are once again as wide as they were thirty years ago, and getting wider, even as college has become far more crucial to lifetime fortunes.

—TAMARA DRAUT, "The Growing College Gap"

You might think that the "they say" here is embedded in the third sentence: They say (or we all think) that a four-year degree is "the all-but-official ticket to middle-class security," and you might assume that Draut will go on to disagree.

If you read the passage this way, however, you would be mistaken. Draut is not questioning whether a college degree has become "the ticket to middle-class security," but whether most Americans can obtain that ticket, whether college is within the financial reach of most American families. You may have been thrown off by the "but" following the statement that college has become a prerequisite for middle-class security. However, unlike the "though" in Zinzchenko's opening, this "but" does not signal that Draut will be disagreeing with the view she has just summarized, a view that in fact she takes as a given. What Draut disagrees with is that this ticket to middle-class security is still readily available to the middle and working classes.

Were one to imagine Draut in a room talking with others with strong views on this topic, one would need to picture her challenging not those who think college is a ticket to financial security (something she agrees with and takes for granted), but those who think the doors of college are open to anyone willing to put forth the effort to walk through them. The view that Draut is challenging, then, is not summarized in her opening. Instead, she assumes that readers are already so familiar with this view that it need not be stated.

Draut's example suggests that in texts where the central "they say" is not immediately identified, you have to construct it yourself based on the clues the text provides. You have to start by locating the writer's thesis and then imagine some of the arguments that might be made against it. What would it look like to disagree with this view? In Draut's case, it is relatively easy to construct a counterargument: it is the familiar

faith in the American Dream of equal opportunity when it comes to access to college. Figuring out the counterargument not only reveals what motivated Draut as a writer but helps you respond to her essay as an active, critical reader. Constructing this counterargument can also help you recognize how Draut challenges your own views, questioning opinions that you previously took for granted.

**WHEN THE "THEY SAY" IS ABOUT SOMETHING
"NOBODY HAS TALKED ABOUT"**

Another challenge in reading for the conversation is that writers sometimes build their arguments by responding to a *lack* of discussion. These writers build their case not by playing off views that can be identified (like faith in the American Dream or the idea that we are responsible for our body weight), but by pointing to something others have overlooked. As the writing theorists John M. Swales and Christine B. Feak point out, one effective way to "create a research space" and "establish a niche" in the academic world is "by indicating a gap in . . . previous research." Much research in the sciences and humanities takes this "Nobody has noticed X" form.

In such cases, the writer may be responding to scientists, for example, who have overlooked an obscure plant that offers insights into global warming, or to literary critics who have been so busy focusing on the lead character in a play that they have overlooked something important about the minor characters.

READING PARTICULARLY CHALLENGING TEXTS

Sometimes it is difficult to figure out the views that writers are responding to not because these writers do not identify

those views but because their language and the concepts they are dealing with are particularly challenging. Consider, for instance, the first two sentences of *Gender Trouble: Feminism and the Subversion of Identity*, a book by the feminist philosopher and literary theorist Judith Butler, thought by many to be a particularly difficult academic writer.

Contemporary feminist debates over the meaning of gender lead time and again to a certain sense of trouble, as if the indeterminacy of gender might eventually culminate in the failure of feminism. Perhaps trouble need not carry such a negative valence.

—JUDITH BUTLER, *Gender Trouble:
Feminism and the Subversion of Identity*

There are many reasons readers may stumble over this relatively short passage, not the least of which is that Butler does not explicitly indicate where her own view begins and the view she is responding to ends. Unlike Zinczenko, Butler does not use the first-person "I" or a phrase such as "in my own view" to show that the position in the second sentence is her own. Nor does Butler offer a clear transition such as "but" or "however" at the start of the second sentence to indicate, as Zinczenko does with "though," that in the second sentence she is questioning the argument she has summarized in the first. And finally, like many academic writers, Butler uses abstract, unfamiliar words that many readers may need to look up, like "gender" (sexual identity, male or female), "indeterminacy" (the quality of being impossible to define or pin down), "culminate" (finally result in), and "negative valence" (a term borrowed from chemistry, roughly denoting "negative significance" or "meaning"). For all these reasons, we can imagine many read-

ers feeling intimidated before they reach the third sentence of Butler's book.

But readers who break down this passage into its essential parts will find that it is actually a lucid piece of writing that conforms to the classic "they say / I say" pattern. Though it can be difficult to spot the clashing arguments in the two sentences, close analysis reveals that the first sentence offers a way of looking at a certain type of "trouble" in the realm of feminist politics that is being challenged in the second.

To understand difficult passages of this kind, you need to translate them into your own words—to build a bridge, in effect, between the passage's unfamiliar terms and ones more familiar to you. Building such a bridge should help you connect what you already know to what the author is saying—and will then help you move from reading to writing, providing you with some of the language you will need to summarize the text. One major challenge in translating the author's words into your own, however, is to stay true to what the author is actually saying, avoiding what we call "the closest cliché syndrome," in which one mistakes a commonplace idea for an author's more complex one (mistaking Butler's critique of the concept of "woman," for instance, for the common idea that women must have equal rights). The work of complex writers like Butler, who frequently challenge conventional thinking, cannot always be collapsed into the types of ideas most of us are already familiar with. Therefore, when you translate, do not try to fit the ideas of such writers into your preexisting beliefs, but instead allow your own views to be challenged. In building a bridge to the writers you read, it is often necessary to meet those writers more than halfway.

For more on
the closest
cliché
syndrome,
see Chapter 2.

So what, then, does Butler's opening say? Translating Butler's words into terms that are easier to understand, we can

see that the first sentence says that for many feminists today, "the indeterminacy of gender"—the inability to define the essence of sexual identity—spells the end of feminism; that for many feminists the inability to define "gender," presumably the building block of the feminist movement, means serious "trouble" for feminist politics. In contrast, the second sentence suggests that this same "trouble" need not be thought of in such "negative" terms, that the inability to define femininity, or "gender trouble" as Butler calls it in her book's title, may not be such a bad thing—and, as she goes on to argue in the pages that follow, may even be something that feminist activists can profit from. In other words, Butler suggests, highlighting uncertainties about masculinity and femininity can be a powerful feminist tool.

Pulling all these inferences together, then, the opening sentences can be translated as follows: "While many contemporary feminists believe that uncertainty about what it means to be a woman will undermine feminist politics, I, Judith Butler, believe that this uncertainty can actually help strengthen feminist politics." Translating Butler's point into our own book's basic move: "They say that if we cannot define 'woman,' feminism is in big trouble. But I say that this type of trouble is precisely what feminism needs." Despite its difficulty, then, we hope you agree that this initially intimidating passage does make sense if you stay with it.

We hope it is clear that critical reading is a two-way street. It is just as much about being open to the way that writers can challenge you, maybe even transform you, as it is about questioning those writers. And if you translate a writer's argument into your own words as you read, you should allow the text to take you outside the ideas that you already hold and to introduce you to new terms and concepts. Even if you end up dis-

agreeing with an author, you first have to show that you have really listened to what he or she is saying, have fully grasped his or her arguments, and can accurately summarize those arguments. Without such deep, attentive listening, any critique you make will be superficial and decidedly *uncritical*. It will be a critique that says more about you than about the writer or idea you're supposedly responding to.

In this chapter we have tried to show that reading for the conversation means looking not just for the thesis of a text in isolation but for the view or views that motivate that thesis—the “they say.” We have also tried to show that reading for the conversation means being alert for the different strategies writers use to engage the view(s) that are motivating them, since not all writers engage other perspectives in the same way. Some writers explicitly identify and summarize a view they are responding to at the outset of their text and then return to it frequently as their text unfolds. Some refer only obliquely to a view that is motivating them, assuming that readers will be able to reconstruct that view on their own. Other writers may not explicitly distinguish their own view from the views they are questioning in ways that all of us find clear, leaving some readers to wonder whether a given view is the writer's own or one that he or she is challenging. And some writers push off against the “they say” that is motivating them in a challenging academic language that requires readers to translate what they are saying into more accessible, everyday terms. In sum, then, though most persuasive writers do follow a conversational “they say / I say” pattern, they do so in a great variety of ways. What this means for readers is that they need to be armed with various strategies for detecting the conversations in what they read, even when those conversations are not self-evident.

THIRTEEN

"THE DATA SUGGEST"

Writing in the Sciences

CHRISTOPHER GILLEN



CHARLES DARWIN DESCRIBED *On the Origin of Species* as "one long argument." In *Dialogue Concerning the Two Chief World Systems*, Galileo Galilei cast his argument for a sun-centered solar system as a series of conversations. As these historical examples show, scientific writing is fundamentally argumentative. Like all academic writers, scientists make and defend claims. They address disagreements and explore unanswered questions. They propose novel mechanisms and new theories. And they advance certain explanations and reject others. Though their vocabulary may be more technical and their emphasis more numerical, science writers use the same

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rhetorical moves as other academic writers. Consider the following example from a 2006 book about the laws of physics.

The common refrain that is heard in elementary discussions of quantum mechanics is that a physical object is in some sense both a wave and a particle, with its wave nature apparent when you measure a wave property such as wavelength, and its particle nature apparent when you measure a particle property such as position. But this is, at best, misleading and, at worst, wrong.

V. J. STENGER, *The Comprehensible Cosmos*, 2006

The “they say / I say” structure of this passage is unmistakable: They say that objects have properties of both waves and particles; I say they are wrong. This example is not a lonely argumentative passage cherry-picked from an otherwise nonargumentative text. Rather, Stenger’s entire book makes the argument that is foreshadowed by its title, *The Comprehensible Cosmos*: that although some might see the universe as hopelessly complex, it is essentially understandable.

Here’s another argumentative passage, this one from a 2001 research article about the role of lactic acid in muscle fatigue:

In contrast to the often suggested role for acidosis as a cause of muscle fatigue, it is shown that in muscles where force was depressed by high $[K^+]_o$, acidification by lactic acid produced a pronounced recovery of force.

O. B. NIELSEN, F. DE PAOLI, AND K. OVERGAARD,
“Protective Effects of Lactic Acid on Force Production in
Rat Skeletal Muscle,” *The Journal of Physiology*, 2001

In other words: Many scientists think that lactic acid causes muscle fatigue, but our evidence shows that it actually promotes recovery. Notice that the authors frame their claim with a version of the “they say / I say” formula: Although

previous work suggests _____, our data argue _____. This basic move and its many variations are widespread in scientific writing. The essential argumentative moves taught in this book transcend disciplines, and the sciences are no exception. The examples in this chapter were written by professional scientists, but they show moves that are appropriate in any writing that addresses scientific issues.

Despite the importance of argument in scientific writing, newcomers to the genre often see it solely as a means for communicating uncontroversial, objective facts. It's easy to see how this view arises. The objective tone of scientific writing can obscure its argumentative nature, and many textbooks reinforce a nonargumentative vision of science when they focus on accepted conclusions and ignore ongoing controversies. And because science writers base their arguments on empirical data, a good portion of many scientific texts *does* serve the purpose of delivering uncontested facts.

However, scientific writing often does more than just report facts. Data are crucial to scientific argumentation, but they are by no means the end of the story. Given important new data, scientists assess their quality, draw conclusions from them, and ponder their implications. They synthesize the new data with existing information, propose novel theories, and design the next experiments. In short, scientific progress depends on the insight and creativity that scientists bring to their data. The thrill of doing science, and writing about it, comes from the ongoing struggle to use data to better understand our world.

START WITH THE DATA

Data are the fundamental currency of scientific argument. Scientists develop hypotheses from existing data and then test

those by comparing their predictions to new experimental data. Summarizing data is therefore a basic move in science writing. Because data can often be interpreted in different ways, describing the data opens the door to critical analysis, creating opportunities to critique previous interpretations and develop new ones.

Describing data requires more than simply reporting numbers and conclusions. Rather than jumping straight to the punch line—to what X concluded—it is important first to describe the hypotheses, methods, and results that led to the conclusion: “To test the hypothesis that _____, X measured _____ and found that _____. Therefore, X concluded _____.” In the following sections, we explore the three key rhetorical moves for describing the data that underpin a scientific argument: presenting the prevailing theories, explaining methodologies, and summarizing findings.

See how a physicist begins with data on pp. 206–13.

Present the Prevailing Theories

Readers must understand the prevailing theories that a study responds to before they can fully appreciate the details. So before diving into specifics, place the work in context by describing the prevailing theories and hypotheses. In the following passage from a 2004 journal article about insect respiration, the authors discuss an explanation for discontinuous gas exchange (DGC), a phenomenon where insects periodically close valves on their breathing tubes.

Lighton (1996, 1998; see also Lighton and Berrigan, 1995) noted the prevalence of DGC in fossorial insects, which inhabit microclimates where CO₂ levels may be relatively high. Consequently, Lighton proposed the chthonic hypothesis, which suggests that

DGC originated as a mechanism to improve gas exchange while at the same time minimizing respiratory water loss.

A. G. GIBBS AND R. A. JOHNSON, "The Role of Discontinuous Gas Exchange in Insects: The Chthonic Hypothesis Does Not Hold Water," *The Journal of Experimental Biology*, 2004

Notice that Gibbs and Johnson not only describe Lighton's hypothesis but also recap the evidence that supports it. By presenting this evidence, Gibbs and Johnson set the stage for engaging with Lighton's ideas. For example, they might question the chthonic hypothesis by pointing out shortcomings of the data or flaws in its interpretation. Or they might suggest new approaches that could verify the hypothesis. The point is that by incorporating a discussion of experimental findings into their summary of Lighton's hypothesis, Gibbs and Johnson open the door to a conversation with Lighton.

Here are some templates for presenting the data that underpin prevailing explanations:

- ▶ Experiments showing _____ and _____ have led scientists to propose _____.
- ▶ Although most scientists attribute _____ to _____, X's result _____ leads to the possibility that _____.

Explain the Methods

Even as we've argued that scientific arguments hinge on data, it's important to note that the quality of data varies depending on how they were collected. Data obtained with sloppy techniques or poorly designed experiments could lead to faulty conclusions. Therefore, it's crucial to explain the methods used to collect data. In order for readers to evaluate a method, you'll need to indicate

its purpose, as the following passage from a journal article about the evolution of bird digestive systems demonstrates:

To test the hypothesis that flowerpiercers have converged with hummingbirds in digestive traits, we compared the activity of intestinal enzymes and the gut nominal area of cinnamon-bellied flowerpiercers (*Diglossa baritula*) with those of eleven hummingbird species.

J. E. SCHONDUBE AND C. MARTINEZ DEL RIO,
Journal of Comparative Physiology, 2004

You need to indicate purpose whether describing your own work or that of others. Here are a couple of templates for doing so:

- ▶ Smith and colleagues evaluated _____ to determine whether _____.
- ▶ Because _____ does not account for _____, we instead used _____.

Summarize the Findings

Scientific data often come in the form of numbers. Your task when presenting numerical data is to provide the context readers need to understand the numbers—by giving supporting information and making comparisons. In the following passage from a book about the interaction between organisms and their environments, Turner uses numerical data to support an argument about the role of the sun's energy on Earth.

The potential rate of energy transfer from the Sun to Earth is prodigious—about 600 W m^{-2} , averaged throughout the year. Of this, only a relatively small fraction, on the order of 1–2 percent, is captured by green plants. The rest, if it is not reflected back into space, is available to do other things. The excess can be considerable: although some natural surfaces reflect as much as 95% of the

incoming solar beam, many natural surfaces reflect much less (Table 3.2), on average about 15–20 percent. The remaining absorbed energy is then capable of doing work, like heating up surfaces, moving water and air masses around to drive weather and climate, evaporating water, and so forth.

J. S. TURNER, *The Extended Organism*, 2000

Turner supports his point that a huge amount of the sun's energy is directly converted to work on Earth by quoting an actual value (600) with units of measurement (W m^{-2} , watts per square meter). Readers need the units to evaluate the value; 600 watts per square inch is very different from 600 W m^{-2} . Turner then makes comparisons using percent values, saying that only 1 to 2 percent of the total energy that reaches Earth is trapped by plants. Finally, Turner describes the data's variability by reporting comparisons as ranges—1 to 2 percent and 15 to 20 percent—rather than single values.

Supporting information—such as units of measurement, sample size (n), and amount of variability—helps readers assess the data. In general, the reliability of data improves as its sample size increases and its variability decreases. Supporting information can be concisely presented as:

- _____ $\pm$ _____ (mean $\pm$ variability) _____ (units),
 $n =$ _____ (sample size).

For example: Before training, resting heart rate of the subjects was 56 ± 7 beats per minute, $n = 12$. Here's another way to give supporting information:

- We measured _____ (sample size) subjects, and the average response was _____ (mean with units) with a range of _____ (lower value) to _____ (upper value).

To help readers understand the data, make comparisons with values from the same study or from other similar work.

Here are some templates for making comparisons:

- ▶ Before training, average running speed was _____ $\pm$ _____ kilometers per hour, _____ kilometers per hour slower than running speed after training.
- ▶ We found athletes' heart rates to be _____ $\pm$ _____ % lower than nonathletes'.
- ▶ The subjects in X's study completed the maze in _____ $\pm$ _____ seconds, _____ seconds slower than those in Y's study.

You will sometimes need to present qualitative data, such as that found in some images and photographs, that cannot be reduced to numbers. Qualitative data must be described precisely with words. In the passage below from a review article about connections between cellular protein localization and cell growth, the author describes the exact locations of three proteins: Scrib, Dlg, and Lgl.

Epithelial cells accumulate different proteins on their apical (top) and basolateral (bottom) surfaces. . . . Scrib and Dlg are localized at the septate junctions along the lateral cell surface, whereas Lgl coats vesicles that are found both in the cytoplasm and "docked" at the lateral surface of the cell.

M. PEIFER, "Travel Bulletin—Traffic Jams Cause Tumors," *Science*, 2000

EXPLAIN WHAT THE DATA MEAN

Once you summarize experiments and results, you need to say what the data mean. Consider the following passage from a study in which scientists fertilized plots of tropical rainforest with nitrogen (N) and/or phosphorus (P).

Although our data suggest that the mechanisms driving the observed respiratory responses to increased N and P may be different, the large CO₂ losses stimulated by N and P fertilization suggest that knowledge of such patterns and their effects on soil CO₂ efflux is critical for understanding the role of tropical forests in a rapidly changing global C [carbon] cycle.

C. C. CLEVELAND AND A. R. TOWNSEND, "Nutrient Additions to a Tropical Rain Forest Drive Substantial Soil Carbon Dioxide Losses to the Atmosphere," *Proceedings of the National Academy of Sciences*, 2006

Notice that in discussing the implications of their data, Cleveland and Townsend use language—including the verbs "suggest" and "may be"—that denotes their level of confidence.

Whether you are summarizing what others say about their data or offering your own interpretation, pay attention to the verbs that connect data to interpretations.

To signify a moderate level of confidence:

- The data *suggest/hint/imply* _____.

To express a greater degree of certainty:

- Our results *show/demonstrate* _____.

Almost never will you use the verb “prove” in reference to a single study, because even very powerful evidence generally falls short of proof unless other studies support the same conclusion.

Scientific consensus arises when multiple studies point toward the same conclusion; conversely, contradictions among studies often signal research questions that need further work. For these reasons, you may need to compare one study’s findings to those of another study. Here, too, you’ll need to choose your verbs carefully.

- ▶ Our data *support/confirm/verify* the work of X by showing that _____.
- ▶ By demonstrating _____, X’s work *extends* the findings of Y.
- ▶ The results of X *contradict/refute* Y’s conclusion that _____.
- ▶ X’s findings *call into question* the widely accepted theory that _____.
- ▶ Our data *are consistent with* X’s hypothesis that _____.

MAKE YOUR OWN ARGUMENTS

Now we turn toward the part of scientific writing where you express your own opinions. One challenge is that the statements of other scientists about their methods and results usually must be accepted. You probably can’t argue, for example, that “X and Y claim to have studied 6 elephants, but I think they actually only studied 4.” However, it might be fair to say, “X and Y studied only 6 elephants, and this small sample size casts doubts on their conclusions.” The second statement doesn’t question what the scientists did or found but instead examines how the findings are interpreted.

When developing your own arguments—the “I say”—you will often start by assessing the interpretations of other scientists. Consider the following example from a review article about the beneficial acclimation hypothesis (BAH), the idea that organisms exposed to a particular environment become better suited to that environment than unexposed animals.

To the surprise of most physiologists, all empirical examinations of the BAH have rejected its generality. However, we suggest that these examinations are neither direct nor complete tests of the functional benefit of acclimation.

R. S. WILSON AND C. E. FRANKLIN, “Testing the Beneficial Acclimation Hypothesis,” *Trends in Ecology & Evolution*, 2002

For more on
the “twist it”
move, see
p. 60.

Wilson and Franklin use a version of the “twist it” move: They acknowledge the data collected by other physiologists but question how those data have been interpreted, creating an opportunity to offer their own interpretation.

You might ask whether we should question how other scientists interpret their own work. Having conducted a study, aren’t they in the best position to evaluate it? Perhaps, but as the above example demonstrates, other scientists might see the work from a different perspective or through more objective eyes. And in fact the culture of science depends on vigorous debate in which scientists defend their own findings and challenge those of others—a give and take that helps improve science’s reliability. So expressing a critical view about someone else’s work is an integral part of the scientific process. Let’s examine some of the basic moves for entering scientific conversations: agreeing, with a difference; disagreeing and explaining why; simultaneously agreeing and disagreeing; anticipating objections; and saying why it matters.

Agree, but with a Difference

Scientific research passes through several levels of critical analysis before being published. Scientists get feedback when they discuss work with colleagues, present findings at conferences, and receive reviews of their manuscripts. So the juiciest debates may have been resolved before publication, and you may find little to disagree with in the published literature of a research field. Yet even if you agree with what you've read, there are still ways to join the conversation—and reasons to do so.

One approach is to suggest that further work should be done:

- ▶ Now that _____ has been established, scientists will likely turn their attention toward _____.
- ▶ X's work leads to the question of _____. Therefore, we investigated _____.
- ▶ To see whether these findings apply to _____, we propose to _____.

Another way to agree and at the same time jump into the conversation is to concur with a finding and then propose a mechanism that explains it. In the following sentence from a review article about dietary deficiencies, the author agrees with a previous finding and offers a probable explanation.

Inadequate dietary intakes of vitamins and minerals are widespread, most likely due to excessive consumption of energy-rich, micronutrient-poor, refined food.

B. AMES, "Low Micronutrient Intake May Accelerate the Degenerative Diseases of Aging through Allocation of Scarce Micronutrients by Triage," *Proceedings of the National Academy of Sciences*, 2006

Here are some templates for explaining an experimental result.

- ▶ One explanation for X's finding of _____ is that _____.
An alternative explanation is _____.
- ▶ The difference between _____ and _____ is probably due to _____.

Disagree—and Explain Why

Although scientific consensus is common, healthy disagreement is not unusual. While measurements conducted by different teams of scientists under the same conditions should produce the same result, scientists often disagree about which techniques are most appropriate, how well an experimental design tests a hypothesis, and how results should be interpreted. To illustrate such disagreement, let's return to the debate about whether or not lactic acid is beneficial during exercise. In the following passage, Lamb and Stephenson are responding to work by Kristensen and colleagues, which argues that lactic acid might be beneficial to resting muscle but not to active muscle.

The argument put forward by Kristensen and colleagues (12) . . . is not valid because it is based on observations made with isolated whole soleus muscles that were stimulated at such a high rate that >60% of the preparation would have rapidly become completely anoxic (4). . . . Furthermore, there is no reason to expect that adding more H^+ to that already being generated by the muscle activity should in any way be advantageous. It is a bit like opening up the carburetor on a car to let in too much air or throwing

gasoline over the engine and then concluding that air and gasoline are deleterious to engine performance.

G. D. LAMB AND D. G. STEPHENSON,

"Point: Lactic Acid Accumulation Is an Advantage during Muscle Activity," *Journal of Applied Physiology*, 2006

Lamb and Stephenson bring experimental detail to bear on their disagreement with Kristensen and colleagues. First, they criticize methodology, arguing that the high muscle stimulation rate used by Kristensen and colleagues created very low oxygen levels (anoxia). They also criticize the logic of the experimental design, arguing that adding more acid (H^+) to a muscle that is already producing it isn't informative. It's also worth noting how they drive home their point, likening Kristensen and colleagues' methodology to flooding an engine with air or gasoline. Even in technical scientific writing, you don't need to set aside your own voice completely.

In considering the work of others, look for instances where the experimental design and methodology fail to adequately test a hypothesis.

- The work of Y and Z appears to show that _____, but their experimental design does not control for _____.

Also, consider the possibility that results do not lead to the stated conclusions.

- While X and Y claim that _____, their finding of _____ actually shows that _____.

Okay, but . . .

Science tends to progress incrementally. New work may refine or extend previous work but doesn't often completely overturn

it. For this reason, science writers frequently agree up to a point and then express some disagreement. In the following example from a commentary about methods for assessing how proteins interact, the authors acknowledge the value of the two-hybrid studies, but they also point out their shortcomings.

The two-hybrid studies that produced the protein interaction map for *D. melanogaster* (12) provide a valuable genome-wide view of protein interactions but have a number of shortcomings (13). Even if the protein-protein interactions were determined with high accuracy, the resulting network would still require careful interpretation to extract its underlying biological meaning. Specifically, the map is a representation of all possible interactions, but one would only expect some fraction to be operating at any given time.

J. J. RICE, A. KERSHENBAUM, AND G. STOLOVITZKY,
 "Lasting Impressions: Motifs in Protein-Protein Maps May
 Provide Footprints of Evolutionary Events." *Proceedings of
 the National Academy of Sciences*, 2005

Delineating the boundaries or limitations of a study is a good way to agree up to a point. Here are some templates for doing so.

- ▶ While X's work clearly demonstrates _____, _____ will be required before we can determine whether _____.
- ▶ Although Y and Z present firm evidence for _____, their data can not be used to argue that _____.
- ▶ In summary, our studies show that _____, but the issue of _____ remains unresolved,

Anticipate Objections

Skepticism is a key ingredient in the scientific process. Before an explanation is accepted, scientists demand convincing evidence and assess whether alternative explanations have been thoroughly explored, so it's essential that scientists consider possible objections to their ideas before presenting them. In the following example from a book about the origin of the universe, Tyson and Goldsmith first admit that some might doubt the existence of the poorly understood "dark matter" that physicists have proposed, and then they go on to respond to the skeptics.

Unrelenting skeptics might compare the dark matter of today with the hypothetical, now defunct "ether," proposed centuries ago as the weightless, transparent medium through which light moved. . . . But dark matter ignorance differs fundamentally from ether ignorance. While ether amounted to a placeholder for our incomplete understanding, the existence of dark matter derives from not from mere presumption but from the observed effects of its gravity on visible matter.

N. D. TYSON AND D. GOLDSMITH, *Origins: Fourteen Billion Years of Cosmic Evolution*, 2004

Anticipating objections in your own writing will help you clarify and address potential criticisms. Consider objections to your overall approach, as well as to specific aspects of your interpretations. Here are some templates for doing so.

- Scientists who take a _____ (reductionist/integrative/biochemical/computational/statistical) approach might view our results differently.

- ▶ This interpretation of the data might be criticized by X, who has argued that _____.
- ▶ Some may argue that this experimental design fails to account for _____.

Say Why It Matters

Though individual studies can be narrowly focused, science ultimately seeks to answer big questions and produce useful technologies. So it's essential when you enter a scientific conversation to say why the work—and your arguments about it—matter. The following passage from a commentary on a research article notes two implications of work that evaluated the shape of electron orbitals.

The classic textbook shape of electron orbitals has now been directly observed. As well as confirming the established theory, this work may be a first step to understanding high-temperature superconductivity.

C. J. HUMPHREYS, "Electrons Seen in Orbit," *Nature*, 1999

Humphreys argues that the study confirms an established theory and that it may lead to better understanding in another area. When thinking about the broad significance of a study, consider both the practical applications and the impact on future scientific work.

- ▶ These results open the door to studies that _____.
- ▶ The methodologies developed by X will be useful for _____.

- ▶ Our findings are the first step toward _____.
- ▶ Further work in this area may lead to the development of _____.

READING AS A WAY OF ENTERING THE SCIENTIFIC CONVERSATION

In science, as in other disciplines, you'll often start with work done by others, and therefore you will need to critically evaluate their work. To that end, you'll need to probe how well their data support their interpretations. Doing so will lead you toward your own interpretations—your ticket into an ongoing scientific conversation. Here are some questions that will help you read and respond to scientific research.

How well do the methods test the hypothesis?

- ▶ Is the sample size adequate?
- ▶ Is the experimental design valid?
Were the proper controls performed?
- ▶ What are the limitations of the methodology?
- ▶ Are other techniques available?

How fairly have the results been interpreted?

- ▶ How well do the results support the stated conclusion?
- ▶ Has the data's variability been adequately considered?
- ▶ Do other findings verify (or contradict) the conclusion?
- ▶ What other experiments could test the conclusion?

What are the broader implications of the work?

Why does it matter?

- ▶ Can the results be generalized beyond the system that was studied?
- ▶ What are the work's practical implications?
- ▶ What questions arise from the work?
- ▶ Which experiments should be done next?

The examples in this chapter show that scientists do more than simply collect facts; they also interpret those facts and make arguments about their meaning. On the frontiers of science, where we are probing questions that are just beyond our capacity to answer, the data are inevitably incomplete and controversy is to be expected. Writing about science presents the opportunity to add your own arguments to the ongoing discussion.

FOURTEEN

"ANALYZE THIS"

Writing in the Social Sciences

ERIN ACKERMAN



SOCIAL SCIENCE is the study of people—how they behave and relate to one another, and the organizations and institutions that facilitate these interactions. People are complicated, so any study of human behavior is at best partial, taking into account some elements of what people do and why, but not always explaining those actions definitively. As a result, it is the subject of constant conversation and argument.

Consider some of the topics studied in the social sciences: minimum wage laws, violence against women, tobacco regulation, the 2000 election, employment discrimination. Got an opinion on any of these topics? You aren't alone. But in the writing you do as a student of the social sciences, you need to

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write about more than just your opinions. Good writing in the social sciences, as in other academic disciplines, requires that you demonstrate that you have thought about what it is you think. The best way to do that is to bring your views into conversation with those expressed by others and to test what you and others think against a review of data. In other words, you'll need to start with what others say and then present what you say as a response.

Consider the following example from a book about contemporary American political culture:

Claims of deep national division were standard fare after the 2000 elections, and to our knowledge few commentators have publicly challenged them. . . . In sum, contemporary observers of American politics have apparently reached a new consensus around the proposition that old disagreements about economics now pale in comparison to new divisions based on sexuality, morality, and religion, divisions so deep as to justify fears of violence and talk of war in describing them.

This short book advocates a contrary thesis: the sentiments expressed in the previously quoted pronouncements of scholars, journalists, and politicians range from simple exaggeration to sheer nonsense. . . . Many of the activists in the political parties and various cause groups do, in fact, hate each other and regard themselves as combatants in a war. But their hatreds and battles are not shared by the great mass of the American people. . . .

MORRIS P. FIORINA, *Culture War?*
The Myth of a Polarized America, 2004

In other words, "they" (journalists, pundits, other political scientists) say that the American public is deeply divided, whereas Fiorina replies that they have misinterpreted the evidence—

specifically, that they have generalized from a few exceptional cases (activists). Even the title of the book calls into question an idea held by others, one Fiorina labels a “myth.”

This chapter explores some of the basic moves social science writers make. In addition, writing in the social sciences generally includes several core components: a strong introduction and thesis, a literature review, and the writer’s own analysis, including presentation of data and consideration of implications. Much of your own writing will include one or more of these components as well. The introduction sets out the thesis, or point, of the paper, briefly explaining what you will say in your text and how it fits into the preexisting conversation. The literature review summarizes what has already been said on your topic. Your analysis allows you to present data—the information about human behavior you are measuring or testing against what other people have said—and to explain the conclusions you have drawn based on your investigation. Do you agree, disagree, or some combination of both, with what has been said by others? What reasons can you give for why you feel that way? And so what? Who should be interested in what you have to say, and why?

THE INTRODUCTION AND THESIS: “THIS PAPER CHALLENGES . . .”

Your introduction sets forth what you plan to say in your essay. You might evaluate the work of earlier scholars or certain widely held assumptions and find them incorrect when measured against new data. Alternatively, you might point out that an author’s work is largely correct, but that it could use some qualifications or be extended in some way. Or you might identify a gap in our knowledge—we know a great deal about topic

X but almost nothing about some other closely related topic. In each of these instances, your introduction needs to cover both "they say" and "I say" perspectives. If you stop after the "they say," your readers won't know what you are bringing to the conversation. Similarly, if you were to jump right to the "I say" portion of your argument, readers might wonder why you need to say anything at all.

Sometimes you join the conversation at a point where the discussion seems settled. One or more views about a topic have become so widely accepted among a group of scholars or society at large that these views are essentially the conventional way of thinking about the topic. You may wish to offer new reasons to support this interpretation, or you may wish to call these standard views into question. To do so, you must first introduce and identify these widely held beliefs and then present your own view. In fact, much of the writing in the social sciences takes the form of calling into question that which we think we already know. Consider the following example from a 2001 article from *The Journal of Economics Perspectives*:

Fifteen years ago, Milton Friedman's 1957 treatise *A Theory of the Consumption Function* seemed badly dated. Dynamic optimization theory had not been employed much in economics when Friedman wrote, and utility theory was still comparatively primitive, so his statement of the "permanent income hypothesis" never actually specified a formal mathematical model of behavior derived explicitly from utility maximization . . . [W]hen other economists subsequently found multiperiod maximizing models that could be solved explicitly, the implications of those models differed sharply from Friedman's intuitive description of his "model." Furthermore, empirical tests in the 1970s and 1980s often rejected these rigorous versions of the permanent income hypothesis in favor of an

alternative hypothesis that many households simply spent all of their current income.

Today, with the benefit of a further round of mathematical (and computational) advances, Friedman's (1957) original analysis looks more prescient than primitive . . .

CHRISTOPHER D. CARROLL, "A Theory of Consumption Function, With and Without Liquidity Constraints,"
The Journal of Economic Perspectives, 2001

This introduction makes clear that Carroll will defend Milton Friedman against some major criticisms of his work. Carroll mentions what has been said about Friedman's work and then goes on to say that the critiques turn out to be wrong and to suggest that Friedman's work reemerges as persuasive. A template of Carroll's introduction might look something like this: Economics research in the last fifteen years suggested Friedman's 1957 treatise was _____ because _____. In other words, they say that Friedman's work is not accurate because of _____, _____, and _____. Recent research convinces me, however, that Friedman's work makes sense.

In some cases, however, there may not be a strong consensus among experts on a topic. You might enter the ongoing debate by casting your vote with one side or another or by offering an alternative view. In the following example, Shari Berman identifies two competing accounts of how to explain world events in the twentieth century and then puts forth a third view.

Conventional wisdom about twentieth-century ideologies rests on two simple narratives. One focuses on the struggle for dominance between democracy and its alternatives. . . . The other narrative focuses on the competition between free-market capitalism and its rivals. . . . Both of these narratives obviously contain some truth. . . . Yet both only tell part of the story, which is why their common

conclusion—neoliberalism as the “end of History”—is unsatisfying and misleading.

What the two conventional narratives fail to mention is that a third struggle was also going on: between those ideologies that believed in the primacy of economics and those that believed in the primacy of politics.

SHARI BERMAN, “The Primacy of Economics versus the Primacy of Politics: Understanding the Ideological Dynamics of the Twentieth Century,” *Perspectives on Politics*, 2009

After identifying the two competing narratives, Berman suggests a third view—and later goes on to argue that this third view explains current debates over globalization. A template for this type of introduction might look something like this: In recent discussions of _____, a controversial aspect has been _____. On the one hand, some argue that _____. On the other hand, others argue that _____. Neither of these arguments, however, considers the alternative view that _____.

Given the complexity of many of the issues studied in the social sciences, however, you may sometimes agree *and* disagree with existing views—pointing out things that you believe are

For more on
different ways
of responding,
see Chapter 4.

correct or have merit, while disagreeing with or refining other points. In the example below, anthropologist Sally Engle Merry agrees with another scholar about something that is a key trait of modern society but argues that this trait has a different origin than the other author identifies.

Although I agree with Rose that an increasing emphasis on governing the soul is characteristic of modern society, I see the trans-

formation not as evolutionary but as the product of social mobilization and political struggle.

SALLY ENGLE MERRY, "Rights, Religion, and Community: Approaches to Violence against Women in the Context of Globalization," *Law and Society Review*, 2001

Here are some templates for agreeing and disagreeing:

- ▶ Although I agree with X up to a point, I cannot accept his overall conclusion that _____.
- ▶ Although I disagree with X on _____ and _____, I agree with her conclusion that _____.
- ▶ Political scientists studying _____ have argued that it is caused by _____. While _____ contributes to the problem, _____ is also an important factor.

In the process of examining people from different angles, social scientists sometimes identify gaps—areas that have not been explored in previous research. In a 1998 article on African American neighborhoods, sociologist Mary Pattillo identifies such a gap.

The research on African Americans is dominated by inquiries into the lives of the black poor. Contemporary ethnographies and journalistic descriptions have thoroughly described deviance, gangs, drugs, intergender relations and sexuality, stymied aspiration, and family patterns in poor neighborhoods (Dash 1989; Hagedorn 1988; Kotlowitz 1991; Lemann 1991; MacLeod 1995; Sullivan 1989; Williams 1989). Yet, the majority of African Americans are not

poor (Billingsley 1992). A significant part of the black experience, namely that of working and middle-class blacks, remains unexplored. We have little information about what black middle-class neighborhoods look like and how social life is organized within them. . . . this article begins to fill this empirical and theoretical gap using ethnographic data collected in Groveland, a middle-class black neighborhood in Chicago.

MARY E. PATTILLO, "Sweet Mothers and Gangbangers:
Managing Crime in a Black Middle-Class
Neighborhood," *Social Forces*, 1998

Pattillo explains that much has been said about poor African American neighborhoods. But, she says, we have little information about the experience of working-class and middle-class black neighborhoods—a gap that her article will address.

Here are some templates for introducing gaps in the existing research:

- ▶ Studies of X have indicated _____. It is not clear, however, that this conclusion applies to _____.
- ▶ _____ often take for granted that _____. Few have investigated this assumption, however.
- ▶ X's work tells us a great deal about _____. Can this work be generalized to _____?

Again, a good introduction indicates what you have to say in the larger context of what others have said. Throughout the rest of your paper, you will move back and forth between the "they say" and the "I say," adding more details.

**THE LITERATURE REVIEW:
"PRIOR RESEARCH INDICATES . . ."**

In the literature review, you explain what "they say" in more detail, summarizing, paraphrasing, or quoting the viewpoints to which you are responding. But you need to balance what they are saying with your own focus. You need to characterize someone else's work fairly and accurately but set up the points you yourself want to make by selecting the details that are relevant to your own perspective and observations.

It is common in the social sciences to summarize several arguments at once, identifying their major arguments or findings in a single paragraph.

How do employers in a low-wage labor market respond to an increase in the minimum wage? The prediction from conventional economic theory is unambiguous: a rise in the minimum wage leads perfectly competitive employers to cut employment (George J. Stigler, 1946). Although studies in the 1970's based on aggregate teenage employment rates usually confirmed this prediction, earlier studies based on comparisons of employment at affected and unaffected establishments often did not (e.g., Richard A. Lester, 1960, 1964). Several recent studies that rely on a similar comparative methodology have failed to detect a negative employment effect of higher minimum wages. Analyses of the 1990–1991 increases in the federal minimum wage (Lawrence F. Katz and Krueger, 1992; Card, 1992a) and of an earlier increase in the minimum wage in California (Card, 1992b) find no adverse employment impact.

DAVID CARD AND ALAN KRUEGER,
"Minimum Wages and Employment: A Case Study of the
Fast-Food Industry in New Jersey and Pennsylvania,"
The American Economic Review, 1994;

Card and Krueger cite the key findings and conclusions of works that are relevant to the question they are investigating and the point they plan to address, asking "How do employers in a low-wage labor market respond to an increase in the minimum wage?" They go on, as good writers should, to answer the question they ask. And they do so by reviewing others who have answered that question, noting that this question has been answered in different, sometimes contradictory, ways.

Such summaries are brief, bringing together relevant arguments by several scholars to provide an overview of scholarly work on a particular topic. In writing such a summary, you need to ask yourself how the authors themselves might describe their positions and also consider what in their work is relevant for the point you wish to make. This kind of summary is especially appropriate when you have a large amount of research material on a topic and want to identify the major strands of a debate or to show how the work of one author builds on that of another. Here are some templates for overview summaries:

- ▶ In addressing the question of _____, political scientists have considered several explanations for _____. X argues that _____. According to Y and Z, another plausible explanation is _____.
- ▶ What is the effect of _____ on _____? Previous work on _____ by X and by Y and Z supports _____.

Sometimes you may need to say more about the works you cite. On a midterm or final exam, for example, you may need to demonstrate that you have a deep familiarity with a particular work. And in some disciplines of the social sciences, longer, more detailed literature reviews are the standard. Your instructor and the articles he or she has assigned are your best guides

for the length and level of detail of your literature review. Other times, the work of certain authors is especially important for your argument, and therefore you need to provide more details to explain what these authors have said. See how Martha Derthick summarizes an argument that is central to her 2001 book about the politics of tobacco regulation.

The idea that governments could sue to reclaim health care costs from cigarette manufacturers might be traced to "Cigarettes and Welfare Reform," an article published in the *Emory Law Journal* in 1977 by Donald Gasner, a law professor at the University of Southern Illinois. Garner suggested that state governments could get a cigarette manufacturer to pay the direct medical costs "of looking after patients with smoking diseases." He drew an analogy to the Coal Mine Health and Safety Act of 1969, under which coal mine operators are required to pay certain disability benefits for coal miners suffering from pneumoconiosis, or black lung disease.

MARTHA DERTHICK, *Up In Smoke: From Legislation to Litigation in Tobacco Politics*, 2005

Note that Derthick identifies the argument she is summarizing, quoting its author directly and then adding details about a precedent for the argument.

You may want to include direct quotations of what others have said, as Derthick does. Using an author's exact words helps you demonstrate that you are representing him or her fairly. But you cannot simply insert a quotation; you need to explain to your readers what it means for your point. Consider the following example drawn from a 2004 political science book on the debate over tort reform.

The essence of *agenda setting* was well enunciated by E. E. Schattschneider: "In politics as in everything else, it makes a great

difference whose game we play" (1960, 47). In short, the ability to define or control the rules, terms, or perceived options in a contest over policy greatly affects the prospects for winning."

WILLIAM HALTOM AND MICHAEL MCCANN, *Distorting the Law: Politics, Media, and the Litigation Crisis*, 2004

Notice how Haltom and McCann first quote Schattschneider and then explain in their own words how political agenda setting can be thought of as a game, with winners and losers.

Remember that whenever you summarize, quote, or paraphrase the work of others, credit must be given in the form of a citation to the original work. The words may be your own, but if the idea comes from someone else you must give credit to the original work. There are several formats for documenting sources. Consult your instructor for help choosing which citation style to use.

THE ANALYSIS

The literature review covers what others have said on your topic. The analysis allows you to present and support your own response. In the introduction you indicate whether you agree, disagree, or some combination of both with what others have said. You will want to expand on how you have formed your opinion and why others should care about your topic.

"The Data Indicate . . ."

The social sciences use data to develop and test explanations. Data can be quantitative or qualitative and can come from a number of sources. You might use statistics related to GDP growth, unemployment, voting rates, or demographics. Or you could use surveys, interviews, or other first-person accounts.

Regardless of the type of data used, it is important to do three things: define your data, indicate where you got the data, and then say what you have done with your data. In a 2005 journal article, political scientist Joshua C. Wilson examines a court case about protests at an abortion clinic and asks whether each side of the conflict acts in a way consistent with their general views on freedom of speech.

[T]his paper relies on close readings of in-person, semi-structured interviews with the participants involved in the real controversy that was the *Williams* case.

Thirteen interviews ranging in length from 40 minutes to 1 hour and 50 minutes were conducted for this paper. Of those interviewed, all would be considered "elites" in terms of political psychology/political attitude research—six were active members of Solano Citizens for Life . . . ; two were members of Planned Parenthood Shasta-Diablo management; one was the lawyer who obtained the restraining order, temporary injunction, and permanent injunction for Planned Parenthood; one was the lawyer for the duration of the case for Solano Citizens for life; two were lawyers for Planned Parenthood on appeal; and one was the Superior Court judge who heard arguments for, and finally crafted, the restraining order and injunctions against Solano Citizens for Life. During the course of the interviews, participants were asked a range of questions about their experiences and thoughts in relation to the *Williams* case, as well as their beliefs about the interpretation and limits of the First Amendment right to free speech—both in general, and in relation to the *Williams* case.

JOSHUA C. WILSON. "When Rights Collide:
Anti-Abortion Protests and the Ideological Dilemma
in *Planned Parenthood Shasta-Diablo, Inc. v. Williams*,"
Studies in Law, Politics, and Society, 2005

Wilson identifies and describes his qualitative data—interviews conducted with key parties in the conflict—and explains the nature of the questions he asked.

If your data are quantitative, you will need to explain them similarly. See how political scientist Brian Arbour explains the quantitative data he used to study for a 2009 article in *The Forum* how a change of rules might have affected the outcome of the 2008 Democratic primary contest between Hillary Clinton and Barack Obama.

I evaluate these five concerns about the Democratic system of delegate allocation by "rerunning" the Obama-Clinton contest with a different set of allocation rules, those in effect for the 2008 Republican presidential contest. . . . Republicans allow each state to make their own rules, leading to "a plethora of selection plans" (Shapiro & Bello 2008, 5) . . . To "rerun" the Democratic primary under Republican rules, I need data on the results of the Democratic primary for each state and congressional district and on the Republican delegate allocation rules for each state. The Green Papers (www.thegreenpapers.com), a website that serves as an almanac of election procedures, rules, and results, provides each of these data sources. By "rerunning" the Democratic primaries and caucuses, I use the exact results of each contest.

BRIAN ARBOUR, "Even Closer, Even Longer: What If the 2008 Democratic Primary Used Republican Rules?" *The Forum*, 2009

Note that Arbour identifies his data as primary voting results and the rules for Republican primaries. In the rest of the paper, Arbour shows how his use of these data suggests that political commentators who thought Republican rules would have clarified the close race between Clinton and Obama were wrong and the race would have been "even closer, even longer."

Here are some templates for discussing data:

- ▶ In order to test the hypothesis that _____, we assessed _____. Our calculations suggest _____.
- ▶ I used _____ to investigate _____. The results of this investigation indicate _____.

“But Others May Object . . .”

No matter how strongly your data support your argument, there are almost surely other perspectives (and thus other data) that you need to acknowledge. By considering possible objections to your argument and taking them seriously, you demonstrate that you’ve done your work and that you’re aware of other perspectives—and most important, you present your own argument as part of an ongoing conversation.

See how economist Christopher Carroll acknowledges that there may be objections to his argument about how people allocate their income between consumption and savings.

I have argued here that the modern version of the dynamically optimizing consumption model is able to match many of the important features of the empirical data on consumption and saving behavior. There are, however, several remaining reasons for discomfort with the model.

CHRISTOPHER D. CARROLL, “A Theory of Consumption Function, With and Without Liquidity Constraints,”
The Journal of Economic Perspectives, 2001

Carroll then goes on to identify the possible limitations of his mathematical analysis.

Someone may object because there are related phenomena that your analysis does not explain or because you do not have the right data to investigate a particular question. Or perhaps someone may object to assumptions underlying your argument or how you handled your data. Here are some templates for considering naysayers:

- ▶ _____ might object that _____.
- ▶ Is my claim realistic? I have argued _____, but readers may question _____.
- ▶ My explanation accounts for _____ but does not explain _____. This is because _____.

"Why Should We Care?"

Who should care about your research, and why? Since the social sciences attempt to explain human behavior, it is important to consider how your research affects the assumptions we make about human behavior. In addition, you might offer recommendations for how other social scientists might continue to explore an issue, or what actions policymakers should take.

In the following example, sociologist Devah Pager identifies the implications of her study of the way having a criminal record affects a person applying for jobs.

[I]n terms of policy implications, this research has troubling conclusions. In our frenzy of locking people up, our "crime control" policies may in fact exacerbate the very conditions that lead to crime in the first place. Research consistently shows that finding

quality steady employment is one of the strongest predictors of desistance from crime (Shover 1996; Sampson and Laub 1993; Uggen 2000). The fact that a criminal record severely limits employment opportunities—particularly among blacks—suggests that these individuals are left with few viable alternatives.

DEVAH PAGER, "The Mark of a Criminal Record,"
The American Journal of Sociology, 2003

Pager's conclusion that a criminal record negatively affects employment chances creates a vicious circle, she says: steady employment discourages recidivism, but a criminal record makes it harder to get a job.

In answering the "so what?" question, you need to explain why your readers should care. Although sometimes the implications of your work may be so broad that they would be of interest to almost anyone, it's never a bad idea to identify explicitly any groups of people who will find your work important.

Templates for establishing why your claims matter:

- ▶ X is important because _____
- ▶ Ultimately, what is at stake here is _____
- ▶ The finding that _____ should be of interest to _____ because _____

As noted at the beginning of this chapter, the complexity of people allows us to look at their behavior from many different viewpoints. Much has been, and will be, said about how and why people do the things they do. As a result, we can look at writing in the social sciences as an ongoing conversation.

When you join this conversation, the "they say / I say" framework will help you figure out what has already been said (they say) and what you can add (I say). The components of social science writing presented in this chapter are tools to help you join that conversation.